

Parsing Unrestricted Text into Prosodic Units. A Formal Description

David Casacuberta, Rafael Marín, Lourdes Aguilar

Universitat Autònoma de Barcelona

1 Introduction

Applications in natural language processing and speech technologies such as text-to-speech or recognition systems require text analysis procedures that are able to run in real time with a low computational cost, and that are robust enough to deal with unrestricted input text. Attempts to automatize the processes of obtaining and analyzing data are being made (Hirschberg 1991, Ostendorf and Veilleux 1994); however, to achieve this goal, a model in which units and levels of analysis are defined is needed. This paper¹ presents the axiomatization of a procedure for segmenting texts into prosodic units, suitable to be implemented in a text-to-speech (TTS) system.

In the domain of TTS systems, the interest in applying automatic procedures is steadily expanding, mainly due to the development of multilingual systems in the laboratories and to the growth of vocal services using speech synthesis. Moreover, there is an attempt to overcome one of the main problems in TTS systems: the lack of naturalness due to deficiencies in prosodic and linguistic information. Although it is generally agreed that an appropriate allocation of pauses improves the global quality of a TTS system, it is not clear why this should be so. It could be said that it is chiefly due to the contribution of prosody to speech perception (Nooteboom *et al.* 1978).

Pauses help to segment words because pauses coincide with word boundaries and word groups, because pauses tend to signal semantically and syntactically coherent groups; moreover, pauses provide the human processor with some additional time to cope with linguistic information. Obviously, these perceptual functions become more important in synthetic speech where the signal does not include any acoustic-phonetic redundancy.

However, although we agree on the need for an adequate strategy for assigning pauses, it is not an easy task. On the one hand, a multiplicity of factors is involved in the prosodic segmentation performed by a speaker (Dechert and Raupach 1980); on the other hand, some pauses are optional and are therefore difficult to systematize.

The problem of how to deal with these matters and their application to TTS systems, is the main concern in a growing family of prosodic segmentation algorithms. Assumptions concerning the relation among the linguistic modules of the system influence the form of the algorithm, varying from systems performing a complete syntactic analysis which is then reinterpreted in prosodic terms (Frenkenberger *et al.* 1994) to systems that merely perform a morphological analysis without considering syntactic information (Emeard *et al.* 1992). The most successful approach is nevertheless the one that at the same time considers the importance of both syntactic and non-syntactic factors in the prediction of prosodic boundaries (Bachenko and Fitzpatrick 1990, Hirschberg and Prieto 1996). Following this approach, it is assumed that suprasegmental phenomena cannot be derived exclusively from one source of information.

On the other hand, the theoretical model establishes the units and the levels of analysis that will be used in the algorithm. For instance, Hirschberg and Prieto 1996 adopt Pierrehumbert's theory of intonational description for English (Pierrehumbert 1980) to develop a phrasing module for a TTS system for Mexican Spanish, whereas Bachenko and Fitzpatrick 1990 formalize the psychological model described in Gee and Grosjean 1983.

Within this framework of the study of the relation between disciplines such as syntax and prosody, and the limitations dictated by the application in a TTS system, our study has a three-fold aim. First, we describe a methodology to automatically assign pauses in texts; second, a formal description of this methodology is proposed; and finally, the reliability of the procedure is tested by means of its implementation in a system.

2 The empirical facts

Before presenting the empirical facts to be explained by the formal model, we will devote some space to the units used in this model.

2.1 Units

The stress group (SG) is conceived as a string, formed by zero or more unstressed words preceding a lexically stressed word. For example, the string *el pelo* (“the hair”) is an SG consisting of an unstressed word preceding the stressed one, whereas *ventana* (“window”) is an SG with only one element, the noun *ventana*. Following traditional Spanish linguistic descriptions, the appearance of stress can be related to the lexical category of the word. It is considered that open categories, that is, nouns, verbs, adjectives and adverbs, are lexically stressed words, while closed categories –prepositions, determiners or conjunctions– are unstressed².

A categorial stress group (CSG) can be seen as an SG which includes syntactic properties, that is, a prosodic unit carrying syntactic information.

When labeling an SG to obtain a CSG, it is necessary to differentiate between SGs formed by only a stressed word and SGs with one or more unstressed words preceding the stressed one. As for the first, we simply indicate the syntactic category to which the stressed word belongs. We refer to this category as the head of the CSG. Thus, an SG such as *ventana* is labeled as noun group (ng).

To deal with SGs formed by more than one word, the category of the first unstressed word of the CSG, besides the category of the stressed word, has to be indicated. In this case, the category of the first word will be considered the head of the CSG, and the category of the last word the modifier of the CSG. Going back to the example given above, *el pelo* is a CSG in which the first category is a quantifier³ and the last one a noun. We will abbreviate this as qg(n), where ‘q’ stands for ‘quantifier’, ‘g’ for ‘group’, and ‘n’ for ‘noun’. The letter before ‘g’ refers to the head and the one between brackets to the modifier.

The list of stressed categories and the corresponding CSGs is as follows: adjective (ag), adverb (adg), gerund (g), infinitive (i), noun (ng), participle (pt) and verb (vg).

The list of CSGs formed by more than one word is the following:

- cg: head: conjunction; possible modifiers: a, ad, g, i, n, pt, v.
- ccg: head: coord. conjunction; possible modifiers: a, ad, g, i, n, pt, v.
- clg: head: clitic; possible modifier: v.
- pg: head: preposition; possible modifiers: a, ad, g, i, n, pt, v.
- qg: head: quantifier; possible modifiers: a, ad, pt, v.

We thus have two levels of specification: one which takes into account only the head and one which also specifies the modifier. Consequently, in a sentence such as *Darwin tuvo que reconocer el fracaso de su teoría*, two labelings are possible:

- (1) *[Darwin]ng [tuvo]vg [que reconocer]cg [el fracaso]qg [de su teoría]pg*
Darwin was forced to accept the failure of his theory.
- (2) *[Darwin]ng [tuvo]vg [que reconocer]cg(i) [el fracaso]qg(n) [de su teoría]pg(n)*

In (2), the letter appearing after the name of the CSG indicates its modifier. Thus, for example, the CSG *[de su teoría]pg(n)* is a prepositional group which has a noun as modifier.

2.2 A definition of pause

A pause can be defined as an interruption of the sound stream, which is acoustically marked by a silence and is perceptible to the listener. Since we are dealing with read speech, it is worthwhile to distinguish between orthographically marked and unmarked pauses, optional and mandatory pauses, and reasonable and impossible pauses.

An orthographically marked pause is a pause that is signalled in a text by means of a punctuation sign; any other pause that cannot be related to a textual sign is an orthographically unmarked pause. In this study, we will try to explain and predict the appearance and location of orthographically unmarked pauses. While being aware of the simplification that this implies, we assume that the punctuation signs are associated to a pause.

A pause is considered to be mandatory if the speaker is forced to fragment the sentence, while it is considered to be optional if different speakers could locate the pause in different places in the same sentence.

Finally, we say that a pause is impossible if any Spanish speaker would make such a pause in a neutral reading. On the other hand, if a pause is acceptable to a listener, we are dealing with a reasonable pause.

To achieve a better understanding of these distinctions, it is worth looking at a sentence such as (3):

- (3) *Las silenciosas pausas entre las estaciones dejaron de ser vividas como una suspensión del sentido y ya no fueron vistas como una momentánea parálisis de la norma ni como una gran crisis o relativización.*

Silent pauses between seasons were no longer perceived as an interruption of sense and they were no longer seen as a momentary paralysis of norms, or as a great crisis or as a relativization.

A speaker has read this sentence with the pauses marked in (4):

- (4) *Las silenciosas pausas entre las estaciones dejaron de ser vividas como una suspensión del sentido \$⁴ y ya no fueron vistas como una momentánea parálisis de la norma \$ ni como una gran crisis o relativización.*

It is obvious that mandatory fragmentation must occur in this sentence for the sake of maximum comprehension on the part of the listener. Our speaker has made two pauses, but in fact there is at least one other possible place to locate a reasonable pause: that is, before the verb *dejaron*. On the contrary, a pause before *entre* will generate an unacceptable result.

2.3 Constraints guiding pause location

The analysis of a corpus of read text in Spanish is the source of the description of the prosodic and syntactic factors that guides the pause location in texts (cf. Marín *et al.* 1996). We summarize below the generalizations found in the empirical study which are the basis for the formal model.

The length of the sentence, measured in terms of the number of SGs, determines the obligatory appearance of a pause. This prosodic constraint can be summarized as follows: in a neutral text, at a normal speech rate, at least seven SGs are required to insert a pause, while the existence of ten or

more groups requires the segmentation of the sentence. As a consequence, the pause is optional in a sentence of between seven and nine SGs.

There is also a prosodic explanation of the location of the pause: in general, pauses tend to appear in the middle of a sentence. This fact can be related to the tendency of speakers to balance the length of the resulting phonic groups (cf. Nespor and Vogel 1983).

It is proposed here that the CSG is a unit that can explain the syntactic factors constraining the location of pauses. It has been found that the type of CSG favours or blocks the appearance of a pause before it. According to this, it is possible to establish a hierarchy of CSGs, as follows: $c\bar{c}g > v\bar{g} > c\bar{g} > g\bar{g} > c\bar{l}g > a\bar{d}g > p\bar{g} > q\bar{g} > p\bar{t}g > i\bar{g} > n\bar{g} > a\bar{g}$.

However, the CSG hierarchy is not enough to prevent the impossible pauses. Consider, for example, sequence (5):

(5) [...] *Juan y Pedro dijeron* [...].

John and Peter said.

According to the hierarchy above, a pause will have to appear before *y Pedro*. However, our linguistic competence indicates that this pause is not acceptable: if a pause were to appear in this sentence, it should appear before *dijeron*. In order to solve such cases, a battery of syntactic restriction rules interacts with the hierarchy of CSG. Going back to (5), a rule forbidding the appearance of a pause between two successive nouns is necessary.

3 A formal description

In this section we undertake a formal description of the concepts and methodology outlined in the preceding section. Our aim here is to give a simple description, using the basic elements of set theory, suitable for use in a system running in real time.

3.1 Concepts

A text $\mathbf{T}$ is any n -tuple where if $\mathbf{w} \in \mathbf{T}$ then $\mathbf{w}$ is a Spanish word (token) or a punctuation sign. The set of Spanish words is a primitive in the system.

In any n -tuple, the position of an element $\mathbf{w}$ is defined as the order number that this element has in its n -tuple. So, in the text $\langle \mathbf{de}, \mathbf{la}, \mathbf{casa} \rangle$, $\mathbf{la}$ has the position number two. Evidently, the same word type can appear as different tokens in the same text. In this case, every token has its own position. In the text $\langle \mathbf{de}, \mathbf{la}, \mathbf{casa}, \mathbf{de}, \mathbf{mi}, \mathbf{padre} \rangle$, there are two different token appearances of word-type ‘de’: the first of them has the position number one and the other the position number four.

$\mathbf{tag}$ is a categorial function if and only if, for every $\mathbf{x} \in \mathbf{T}$, where $\mathbf{T}$ is a text, $\mathbf{tag}(\mathbf{x}) = \mathbf{c}$, where $\mathbf{c} \in \mathbf{Cat}$, a finite set that contains all the categories in which Spanish words are divided for the purposes of our study and nothing else: $\mathbf{Cat}$ cannot be the empty set.

For example, let us consider $\mathbf{Cat} = \{\mathbf{noun}, \mathbf{verb}, \mathbf{adjective}, \mathbf{adverb}, \mathbf{quantifier}, \mathbf{preposition}, \mathbf{punctuation-sign}\}$. Then, our function $\mathbf{tag}$, applied to all the elements of $\langle \mathbf{de}, \mathbf{la}, \mathbf{casa} \rangle$, will give us $\mathbf{tag}(\mathbf{de}) = \mathbf{preposition}$, $\mathbf{tag}(\mathbf{la}) = \mathbf{quantifier}$, $\mathbf{tag}(\mathbf{casa}) = \mathbf{noun}$. Depending on the elements that form $\mathbf{Cat}$, we will have different types of functions, with possibly different images for every word type.

If we make some changes to our original $\mathbf{Cat}$ set and develop $\mathbf{Cat}' = \{\mathbf{noun}, \mathbf{verb}, \mathbf{adjective}, \mathbf{adverb}, \mathbf{quantifier}, \mathbf{de-type-preposition}, \mathbf{non-de-type-preposition}, \mathbf{punctuation-sign}\}$, then our new $\mathbf{Cat}'$ will give us $\mathbf{tag}(\mathbf{de}) = \mathbf{de-type-preposition}$. Thus the system is not restricted to an a priori closed group of categories, but rather it offers the possibility of adding, extracting and modifying categories as the study goes along.

Obviously, the function is not sensitive to the position of the element in the utterance, so, using $\mathbf{Cat}$ as the base set, in the utterance $\langle \mathbf{de}, \mathbf{la}, \mathbf{casa}, \mathbf{de}, \mathbf{mi}, \mathbf{padre} \rangle$ both $\mathbf{de}$ in position one and in position four have the same image, that is, $\mathbf{preposition}$, and their different positions have no effect on the result.

Leaving aside the formal aspect, a tagger would be responsible for implementing the function $\mathbf{tag}$ in a computer application.

$\mathbf{st}$ is the stress function when, for every $\mathbf{w} \in \mathbf{T}$, $\mathbf{st}(\mathbf{x}) = \mathbf{s}$, where $\mathbf{s} \in \{\mathbf{stressed}, \mathbf{unstressed}\}$. It is interesting to observe that, contrary to $\mathbf{tag}$, $\mathbf{st}$ is fixed, because stressed and unstressed are basic categories we do not plan to alter; instead, we merely accept phonetic evidence. The $\mathbf{st}$ function can be implemented by means of a database combining phonetic and lexical information to decide whether a word is stressed or not.

A categorial stress group **CSG** from an utterance **U** is an n-tuple, where every element $\mathbf{x} \in \mathbf{CSG}$ has the form $\langle \mathbf{w}, \mathbf{cat}, \mathbf{s} \rangle$, where $\mathbf{w} \in \mathbf{U}$, $\mathbf{Cat} = \mathbf{tag}(\mathbf{w})$ and $\mathbf{s} = \mathbf{st}(\mathbf{w})$ and a) if a **CSG** has n elements, then the element with position number n is a 3-tuple $\langle \mathbf{w}, \mathbf{cat}, \mathbf{s} \rangle$, such that $\mathbf{s} = \mathbf{stressed}$ and if $\langle \mathbf{w}', \mathbf{cat}', \mathbf{s}' \rangle \in \mathbf{CSG}$ and $\langle \mathbf{w}', \mathbf{cat}', \mathbf{s}' \rangle$'s position $< n$, then $\mathbf{s}' = \mathbf{unstressed}$; b) if $\langle \mathbf{w}, \mathbf{cat}, \mathbf{s} \rangle \in \mathbf{CSG}$, $\langle \mathbf{w}, \mathbf{cat}, \mathbf{s} \rangle$'s position in **CSG** is n, $\langle \mathbf{y}, \mathbf{cat}', \mathbf{s}' \rangle \in \mathbf{CSG}$ and $\langle \mathbf{y}, \mathbf{cat}', \mathbf{s}' \rangle$'s position in **CSG** is n+1, then $\mathbf{w} \in \mathbf{U}$, $\mathbf{w}$'s position in **U** is m, $\mathbf{y} \in \mathbf{U}$ and $\mathbf{y}$'s position in **U** is m+1; c) $\mathbf{CSG} \neq \emptyset$.

We will define **group** as a function that for every utterance **U** with m elements gives an n -tuple **TCSG**, where $m \geq n$ and every element $\mathbf{w} \in \mathbf{TCSG}$ is a **CSG** obtained from the utterance **U** and: a) for every $\mathbf{w1} \in \mathbf{U}$ there is one and only one **CSG** $\in \mathbf{TCSG}$ such that $\langle \mathbf{w1}, \mathbf{tag(w1)}, \mathbf{st(w1)} \rangle \in \mathbf{CSG}$; b) for every $\mathbf{w1}, \mathbf{w2} \in \mathbf{U}$ such that $\mathbf{w1}$ position is n and $\mathbf{w2}$ position is $n+1$ then either: 1) there is one **CSG** such that $\langle \mathbf{w1}, \mathbf{tag(w1)}, \mathbf{st(w1)} \rangle \in \mathbf{CSG}$, $\langle \mathbf{w2}, \mathbf{tag(w2)}, \mathbf{st(w2)} \rangle \in \mathbf{CSG}$ and if $\langle \mathbf{w1}, \mathbf{tag(w1)},$

$\text{st}(\mathbf{w1})\rangle$ position is m , then $\langle \mathbf{w2}, \text{tag}(\mathbf{w2}), \text{st}(\mathbf{w2})\rangle$ is $m+1$; or 2) there are $\mathbf{CSG1}, \mathbf{CSG2} \in \text{TCGS}$ such that $\mathbf{CSG1}$ position is x and $\mathbf{CSG2}$ is $x+1$ and the last element of $\mathbf{CSG1}$ is $\langle \mathbf{w1}, \text{tag}(\mathbf{w1}), \text{st}(\mathbf{w1})\rangle$ and the first element of $\mathbf{CSG2}$ is $\langle \mathbf{w2}, \text{tag}(\mathbf{w2}), \text{st}(\mathbf{w2})\rangle$.

Not every utterance $\mathbf{U}$ can be an argument for **group**. The necessary and sufficient condition for it is that the last word of the utterance is stressed. Observe that, if the last word is not stressed, then no matter how the former words are packed into CSGs, this last word cannot form part of a CSG because there are by definition no further elements after the last one. This last word of the utterance should be the end of a CSG, but according to point a) in the definition of CSG (cf. §3.1) the last element of a CSG must be stressed. Since we assumed that the last element was unstressed, this poses a contradiction, and $\mathbf{U}$ cannot be divided into CSGs.

3.2.2 Modeling the prosodic constraints

The first prosodic constraint can be translated as follows: in a text, the more SGs there are, the more likely it will be that a pause is needed. We can assign this criterion by means of presenting an interval of SGs where the pause is possible. Here is the function pause-type: **pt** is a function that goes from $\mathbf{N}$ to **typause**, where **typause** is a set whose members are **typause** = {**non-existent**, **optional**, **mandatory**}. Integers from 1 to 6 have **non-existent** as image, integers from 7 to 9 have **optional** as image, and integers greater than 9 have **mandatory** as image.

The other prosodic constraint states that in a sentence, the more centered the pause, the better. We model it with a function that takes an utterance and erases the first and last three SGs in it. Formally: **cut** is a function that takes an utterance $\mathbf{U1}$ of length m and produces another utterance $\text{cut}(\mathbf{U1}) = \mathbf{U2}$ where $\mathbf{U2}$ length is $m-6$ and: a) $\mathbf{U2}$ is a subsequence of $\mathbf{U1}$, that is, $\mathbf{U1}$ is like a text from which $\mathbf{U2}$ has been obtained; b) if $\text{pt}(\mathbf{U1}) = \text{non-existent}$ then $\text{cut}(\mathbf{U1}) = \mathbf{U1}$. If that is not the case, then: c) the first element of $\mathbf{U2}$ is $\mathbf{w}$, where $\mathbf{w} \in \mathbf{U1}$ and $\mathbf{w}$ position is 4; d) the last element of $\mathbf{U2}$ is $\mathbf{w2}$, where $\mathbf{w2} \in \mathbf{U1}$ and $\mathbf{w2}$ position is $m-3$.

3.2.3 Modeling the syntactic constraints

A syntactic constraint stipulates which categories are more likely to have a pause; formally, the function **catco** is used to model this constraint.

The function **catco** has two arguments, both belonging to **Cat**, which represent the head and the modifier. It has a natural number as image, which indicates the probability of a CSG with the first category of **catco** as head and the second as the modifier having a pause before. When the head and the modifier coincide we simply repeat the category. Extensionally, **catco** can be defined as follows: **catco**(cc,v) = 1; **catco**(cc,n) = 2; **catco**(v,v) = 3; **catco**(c,v) = 4; **catco**(g,g) = 5; **catco**(cl,v) = 6; **catco**(ad,ad) = 7; **catco**(p,n) = 8; **catco**(p,v) = 9; **catco**(p,a) = 10; ...

These probability assignations were obtained by the authors in an empirical study described in Marín *et al.* 1996. Other combinations are in principle possible but only very occasionally present pauses, so they have associated a 0, indicating that no pause can be found.

There are in fact two different **catco**, because optional and mandatory pauses have different probabilities in the assignation of pauses. So, we use **catco2** to determine the most likely place for a pause to appear in a sentence that has an optional pause and **catco1** to calculate the probability of pause in mandatory pause sentences. These functions are alike, except that some of the items are not considered, and therefore assigned numerical probabilities differ.

3.2.4 Restriction rules

As mentioned above (cf. §2.3), the hierarchy of CSG types is not sufficient to prevent some impossible pauses. In these cases, a battery of restriction rules has been defined. Formally, the restriction rules are governed by the function **rest**: **rest** is a function that assigns a ‘yes’ if there is no restriction between two **CSG** types to have a pause and ‘no’ if the pause is inadmissible: **rest**: ($\langle \text{Cat}, \text{Cat} \rangle, \langle \text{Cat}, \text{Cat} \rangle$) $\rightarrow$ **yes**, **no**. We present **rest** extensionally: **rest**($\langle \text{cc}, \text{v} \rangle, \langle \text{n}, \text{n} \rangle$) = **yes**; **rest**($\langle \text{v}, \text{v} \rangle, \langle \text{cc}, \text{n} \rangle$) = **yes**; **rest**($\langle \text{n}, \text{n} \rangle, \langle \text{v}, \text{v} \rangle$) = **yes**; **rest**($\langle \text{a}, \text{a} \rangle, \langle \text{cc}, \text{a} \rangle$) = **no**; **rest**($\langle \text{ad}, \text{ad} \rangle, \langle \text{v}, \text{v} \rangle$) = **no** ...

3.2.5 A model for locating pauses in a text

Our algorithm takes a text and returns an n-tuple where this text is divided into utterances; there should be no pause within these utterances, although there should be a pause between each of them. First, the system takes the text and applies **Cat** and **st** to every word of it. Then, the text is divided into utterances, obtaining an **ftu** of the text with every utterance in it. The

next step is segmenting every utterance into CSGs. Afterwards, with the help of **pt** the system decides whether or not a pause is needed for each utterance. If there is no pause, the task is complete.

However, if there is a mandatory or optional pause, the second prosodic constraint is applied and **cut** is activated. Once the new utterance is obtained, the syntactic constraint is applied by means of **catco**. The possible place for a pause is checked using **rest** to make sure that it is really a suitable place for a pause. If it is not, another position is searched using **catco**, examined with **rest** until a satisfactory place to insert a pause is found. In the case of an optional pause, no other position is searched. Once this is done, the remaining utterances are checked with **pt** until every utterance has been successfully divided into pauses.

4 Evaluating the methodology

The formal procedure, able to capture the empirical facts, has been implemented in Edinburgh standard Prolog. The program designed to assign pauses in unrestricted texts contains a set of rules and a database with the words labeled with their lexical category. The text is read from an ASCII format and the output is another ASCII document that duplicates the text, but marking all the mandatory and reasonable pauses the system computes.

Let us now consider how the program will work on a fragment of text, by means of the following two examples:

- (6) *Trabajaba en Barcelona.*

He/she worked in Barcelona.

- (7) *Su módulo de tres cuerpos y la acertada resolución espacial están a la altura de cualquier modelo europeo y son superiores a los americanos.*

Its three-story module and its successful spatial resolution are comparable to any European model and are superior to the American ones.

The sentence in (6) has two SGs, so it is too short to contain a pause. Since a pause cannot appear if the sentence has less than seven SGs, the program will not break the sentence.

However, the sentence in (7) has fourteen SGs, so according to the first prosodic constraint, a pause will be obligatory. Regarding the location of

the pause, it has been mentioned above that pauses tend to appear in the center of a sentence.

This is implemented by removing the first and the last three SGs in an utterance, as shown when comparing (8a) with (8b):

- (8) a. *[Su módulo] [de tres] [cuerpos] [y la acertada] [resolución] [espacial] [están] [a la altura] [de cualquier] [modelo] [europeo] [y son] [superiores] [a los americanos]*
 b. *[y la acertada] [resolución] [espacial] [están] [a la altura] [de cualquier] [modelo] [europeo]*

The most central place would be between the CSGs *a la altura* and *de cualquier*. However, these two CSGs maintain a strong syntactic dependence, while there is less syntactic cohesion between *espacial* and *están*. This is represented by the hierarchy of CSGs implemented with the function **catco1**: it is preferable to insert the pause before a [vg] rather than before a [pg]. Therefore, the program will locate the pause before *[están]*. After the first pause is located, we have two new utterances:

- (9) a. *Su módulo de tres cuerpos y la acertada resolución espacial*
 b. *están a la altura de cualquier modelo europeo y son superiores a los americanos.*

The first one is only six CSGs long, so it cannot be segmented. The second one is eight CSGs long, so it may have an optional pause. The location of the pause is determined by the hierarchy of CSGs and solved by **catco2**. Once **catco2** is loaded, we find a place where an optional pause is suitable, namely between *europeo* and *y son*, that is before a cc(v). The final result would be as follows:

- (10) *Su módulo de tres cuerpos y la acertada resolución espacial \$ están a la altura de cualquier modelo europeo \$ y son superiores a los americanos.*

If we compare this segmentation with another performed by a speaker, the obligatory pauses are found to coincide and it is only in the domain of optionality that divergences are found.

5 Conclusions

In this paper, a formal description of a syntactico-prosodic analysis for unrestricted text has been presented. The central unit in the analysis is the CSG, an SG which is syntactically labeled; in other words, a prosodic unit which carries syntactic information. This unit has proved to be useful in a system for automatically assigning pauses in texts.

The empirical facts have been formalized and implemented in a system. This syntactico-prosodic system can assign all the mandatory pauses a text needs, and also generate some optional pauses that are always reasonable according to our definition above, the final result being an acceptable assignation of pauses. The formal model avoids syntactic recursion, which makes it possible to have a program with greater computational simplicity, speed and effectiveness. This is especially important in such applications as TTS systems, where all transformations have to be performed in real time.

Notes

1. We would like to thank Sergio Balari for his valuable comments. R. Marín's work was partially supported by a grant from Generalitat de Catalunya (AP-LP/95-2704).
2. For a more detailed discussion of stressed and unstressed words in Spanish, see Navarro Tomás 1925.
3. We use "quantifier" in its broad sense, including categories such as determiners, demonstratives, possessives, ...
4. Throughout the article, we will use the sign '\$' to represent a pause.

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